

$$\Rightarrow (D^2 + 2D + 2)(D^2 + 2 - 2D) = 0$$

$$\Rightarrow D^2 + 2D + 2 = 0,$$

$$\therefore D = \frac{-2 \pm \sqrt{4 - 8}}{2}$$

$$= \frac{-2 \pm \sqrt{-4}}{2}$$

$$= \frac{-2 \pm 2i}{2}$$

$$= -1 \pm i$$

$$(\alpha = -1, \beta = 1)$$

$$D^2 - 2D + 2 = 0$$

$$D = \frac{2 \pm \sqrt{4 - 8}}{2}$$

$$= \frac{2 \pm 2i}{2}$$

$$= 1 \pm i$$

$$(\alpha = 1, \beta = 1)$$

General solution:

$$x = e^{-t} [C_1 \cos t + C_2 \sin t] + e^t [C_3 \cos t + C_4 \sin t]$$

Solve the following diff. eqs.

(1) $\frac{d^4 y}{dx^4} - y = 0$

(2) $(D^2 - 6D + 9)y = 0$

(3) $\frac{d^3 y}{dx^3} - 13 \frac{dy}{dx} + 12y = 0$

(4) $(D^3 + 1)y = 0$

(5) $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 5y = 0$

Solution of non-homogeneous diff. eq.

Consider the diff. eq:

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = \phi(x)$$

$$\therefore (a_0 D^2 + a_1 D + a_2)y = \phi(x) \rightarrow (1)$$

$$\therefore f(D)y = \phi(x)$$

$$\text{where } f(D) = a_0 D^2 + a_1 D + a_2$$

Homogeneous part of this eq is given by

$$(a_0 D^2 + a_1 D + a_2)y \rightarrow (2)$$

$$\therefore f(D)y = 0$$

General solution of eq (2) is called complementary function (CF) and any solution of (1) is called a particular integral (PI)

Theorem

Consider the diff. eq (1)

$$(a_0 D^2 + a_1 D + a_2) y = \phi(x) \longrightarrow (1)$$

It's homogeneous part is given by

$$(a_0 D^2 + a_1 D + a_2) y = 0 \longrightarrow (2)$$

General solution of eq (1) is given by

$$y = CF + PI$$

Proof Let y_g be the general soln of eq (2)

$$\text{i.e. } (a_0 D^2 + a_1 D + a_2) y_g = 0 \longrightarrow (3)$$

and y_p be a particular soln of eq (1)

$$\text{i.e. } (a_0 D^2 + a_1 D + a_2) y_p = \phi(x) \longrightarrow (4)$$

$$\text{Let } y = y_g + y_p \longrightarrow (3)$$

$$\begin{aligned} (a_0 D^2 + a_1 D + a_2) y &= (a_0 D^2 + a_1 D + a_2) (y_g + y_p) \\ &= (a_0 D^2 + a_1 D + a_2) y_g + (a_0 D^2 + a_1 D + a_2) y_p \\ &= 0 + \phi(x) \quad (\text{using (3) and (4)}) \\ &= \phi(x) \end{aligned}$$

This shows that $y = y_g + y_p$ satisfies eq (1)

$\therefore$ It is a solution of eq (1).

Since y_g is the gen. soln of eq (2)

it contains two arbitrary constants

eq (3) gives the gen. soln of (1).

$$\text{i.e. } y = y_g + y_p = CF + PI \text{ is the gen. soln.}$$

Special methods of finding Particular integral

consider the diff. eq

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = \phi(x)$$

$$\text{or } f(D)y = \phi(x) \quad \text{where } f(D) = a_0 D^2 + a_1 D + a_2$$

It's gen. soln is given by

$$y = CF + PI$$

Case (I) $\phi(x) = e^{ax}$

$$D(e^{ax}) = a e^{ax}, \quad D^2(e^{ax}) = a^2 e^{ax}$$

$$\begin{aligned} \therefore f(D)(e^{ax}) &= (a_0 D^2 + a_1 D + a_2) e^{ax} \\ &= a_0 a^2 e^{ax} + a_1 a e^{ax} + a_2 e^{ax} \\ &= (a_0 a^2 + a_1 a + a_2) e^{ax} \\ &= f(a) e^{ax} \end{aligned}$$

Operating both sides by $f(D)^{-1} = \frac{1}{f(D)}$, we get

$$\frac{1}{f(D)} \cdot f(D) e^{ax} = \frac{1}{f(D)} f(a) e^{ax}$$

$$\text{or } e^{ax} = f(a) \cdot \frac{1}{f(D)} e^{ax}$$

$$\therefore \frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax} \quad \text{provided } f(a) \neq 0.$$

$$\text{or } PI = \frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax} \quad \text{if } f(a) \neq 0$$

Note $f(a) = 0 \Rightarrow a$ is a root of the A.E.
 $f(D) = 0$

If 'a' is a root of multiplicity r then

$$PI = \frac{x^r}{f^{(r)}(a)} e^{ax} \quad \text{where } f^{(r)}(a) = f^{(r)}(D) \text{ at } D=a$$

$$\text{when } r=1, \quad PI = \frac{x}{f'(a)} e^{ax}$$

$$\text{when } r=2, \quad PI = \frac{x^2}{f''(a)} e^{ax} \dots$$

Examples

(1) solve $\frac{d^2 y}{dx^2} - 13 \frac{dy}{dx} + 12y = e^{-2x}$

Associated eq is given by

$$\frac{d^2 y}{dx^2} - 13 \frac{dy}{dx} + 12y = 0 \Rightarrow (D^2 - 13D + 12)y = 0$$

A.E. is given by $D^2 - 13D + 12 = 0$

$$\therefore (D-1)(D-12) = 0 \Rightarrow D = 1, 12$$

$$\therefore CF = C_1 e^x + C_2 e^{12x}$$

$$PI = \frac{1}{f(D)} e^{-2x} = \frac{1}{D^2 - 13D + 12} e^{-2x} \quad (a = -2)$$

$$= \frac{1}{(-2)^2 - 13(-2) + 12} e^{-2x} \quad \left(\begin{array}{l} \text{Replacing} \\ D \text{ by } -2 \end{array} \right)$$

$$= \frac{1}{4 + 26 + 12} e^{-2x}$$

$$= \frac{1}{42} e^{-2x}$$

Gen. soln. is given by

$$y = CF + PI = C_1 e^x + C_2 e^{12x} + \frac{1}{42} e^{-2x}$$

(2) solve $\frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} - 10y = e^{2x}$

$$\therefore (D^2 + 3D - 10)y = e^{2x}$$

A.E. is $D^2 + 3D - 10 = 0$

$$\therefore (D-2)(D+5) = 0 \Rightarrow D = 2, -5$$

$$CF = C_1 e^{2x} + C_2 e^{-5x}$$

$$PI = \frac{1}{f(D)} e^{2x} = \frac{1}{D^2 + 3D - 10} e^{2x} \quad \left(\begin{array}{l} a=2 \text{ which is} \\ \text{single root of A.E.} \end{array} \right)$$

$$= \frac{x}{f'(a)} e^{2x}$$

$$= \frac{x}{7} e^{2x}$$

$$\left| \begin{array}{l} f(D) = D^2 + 3D - 10 \\ f'(D) = 2D + 3 \\ f'(a) = 2(2) + 3 = 7 \end{array} \right|$$

$$\text{gen. soln. } y = CF + PI = C_1 e^{2x} + C_2 e^{-5x} + \frac{x}{7} e^{2x}$$

$$(3) \frac{d^2 y}{dx^2} - 12 \frac{dy}{dx} + 36y = e^{6x}$$

$$\text{or } (D^2 - 12D + 36)y = e^{6x}$$

$$\text{A.E. is } D^2 - 12D + 36 = 0$$

$$\text{or } (D-6)^2 = 0 \Rightarrow D = 6, 6$$

$$\text{CF} = (C_1 + C_2 x) e^{6x}$$

$$\text{PI} = \frac{1}{D^2 - 12D + 36} e^{6x}$$

$$= \frac{x^2}{f''(a)} e^{6x}$$

$$= \frac{x^2}{2} e^{6x}$$

($a=6$ which is a double root of the A.E.)

$$\left| \begin{array}{l} f(D) = D^2 - 12D + 36 \\ f'(D) = 2D - 12 \\ f''(D) = 2 \therefore f''(6) = 2 \end{array} \right|$$

gen. soln is $y = \text{CF} + \text{PI}$

$$= (C_1 + C_2 x) e^{6x} + \frac{x^2}{2} e^{6x}$$

Solve (1) $\frac{d^2 y}{dx^2} + 6 \frac{dy}{dx} + 5y = 2e^x + 10e^{-5x}$

(2) $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 4y = 4e^x - 18e^{-x}$

(3) $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = e^x + 4e^{3x}$

Answers

(1) $(D^2 + 6D + 5)y = 2e^x + 10e^{-5x}$

A.E. is $D^2 + 6D + 5 = 0$

or $(D+1)(D+5) = 0 \Rightarrow D = -1, -5$

CF = $C_1 e^{-x} + C_2 e^{-5x}$

PI = $\frac{1}{D^2 + 6D + 5} (2e^x + 10e^{-5x})$

$= \frac{1}{D^2 + 6D + 5} (2e^x) + \frac{1}{D^2 + 6D + 5} (10e^{-5x})$

$= \text{PI}_1 + \text{PI}_2$

PI₁ = $\frac{1}{D^2 + 6D + 5} (2e^x) = \frac{1}{1^2 + 6(1) + 5} (2e^x)$

($a=1$ which is not a root of A.E.)
 $\therefore$ Replace D by 1

$$i \quad PI_1 = \frac{1}{12} (2e^x) = \frac{1}{6} e^x$$

$$PI_2 = \frac{1}{D^2 + bD + 5} (10e^{-5x})$$

$$= \frac{x}{f'(a)} (10e^{-5x}) \quad \left(a = -5 \text{ which is a single root of the A.E.} \right)$$

$$= \frac{x}{-4} (10e^{-5x})$$

$$= -\frac{5x}{2} e^{-5x}$$

$$\left| \begin{array}{l} f(D) = D^2 + bD + 5 \\ f'(D) = 2D + b \\ f'(a) = 2(-5) + b = -4 \end{array} \right|$$

$$\text{gen. soln } y = CF + PI$$

$$y = CF + PI_1 + PI_2$$

$$i.e. \quad y = c_1 e^{\frac{x}{2}} + c_2 e^{-\frac{\sqrt{7}}{2}x} + \frac{1}{6} e^x - \frac{5}{2} x e^{-5x}$$

$$(2) \quad (D^2 - 3D + 4)y = 4e^x - 18e^{-x}$$

$$\text{A.E. is } D^2 - 3D + 4 = 0$$

$$D = \frac{3 \pm \sqrt{9 - 16}}{2} = \frac{3 \pm i\sqrt{7}}{2} \quad \left(\alpha = \frac{3}{2}, \beta = \frac{\sqrt{7}}{2} \right)$$

$$CF = e^{\alpha x} [c_1 \cos \beta x + c_2 \sin \beta x]$$

$$= e^{\frac{3x}{2}} \left[c_1 \cos \frac{\sqrt{7}}{2} x + c_2 \sin \frac{\sqrt{7}}{2} x \right]$$

$$PI = \frac{1}{D^2 - 3D + 4} (4e^x - 18e^{-x})$$

$$= \frac{1}{D^2 - 3D + 4} (4e^x) + \frac{1}{D^2 - 3D + 4} (-18e^{-x})$$

$$= \frac{1}{D^2 - 3D + 4} (4e^x) + \frac{1}{(-1)^2 - 3(-1) + 4} (-18e^{-x})$$

$$= \frac{1}{2} (4e^x) + \frac{1}{8} (-18e^{-x})$$

$$= 2e^x - \frac{9}{2} e^{-x}$$

$$\text{gen. soln } y = CF + PI$$

$$= e^{\frac{3x}{2}} \left[c_1 \cos \frac{\sqrt{7}}{2} x + c_2 \sin \frac{\sqrt{7}}{2} x \right] + 2e^x - \frac{9}{2} e^{-x}$$

$$(3) (D^2 - 3D + 2)y = e^x + 4e^{3x}$$

$$\text{A.E. is } D^2 - 3D + 2 = 0 \Rightarrow (D-1)(D-2) = 0$$

$$\therefore D = 1, 2$$

$$CF = C_1 e^x + C_2 e^{2x}$$

$$PI = \frac{1}{D^2 - 3D + 2} (e^x + 4e^{3x})$$

$$= \frac{1}{D^2 - 3D + 2} (e^x) + \frac{1}{D^2 - 3D + 2} (4e^{3x})$$

$$= PI_1 + PI_2$$

$$PI_1 = \frac{1}{D^2 - 3D + 2} (e^x)$$

(Here $a=1$ which is a single root of the A.E.)

$$= \frac{x}{f'(a)} e^{ax}$$

$$= \frac{x}{-1} e^x = -x e^x$$

$$\left(\begin{array}{l} f(D) = D^2 - 3D + 2 \\ f'(D) = 2D - 3 \\ f'(a) = 2(1) - 3 = -1 \end{array} \right)$$

$$PI_2 = \frac{1}{D^2 - 3D + 2} (4e^{3x}) \quad \left(\text{Here } a=3 \text{ which is not a root of the A.E.} \right)$$

$$= \frac{1}{3^2 - 3(3) + 2} (4e^{3x}) = \frac{1}{2} (4e^{3x}) = 2e^{3x}$$

$$\text{Gen. Soln } y = CF + PI = C_1 e^x + C_2 e^{2x} - x e^x + 2e^{3x}$$

Solve the following diff. eqs

$$(1) (D^2 - 5D + 6)y = e^{5x}$$

$$(2) (D^2 + 5D + 7)y = e^{-x}$$

$$(2) (D^2 + 4D + 4)y = e^{-3x}$$

$$(4) \frac{d^2 y}{dx^2} - 9y = 5e^{4x} + 2e^{3x}$$

$$(5) (D^2 - 6D + 8)y = (e^{2x} - 1)^2$$

Case II when $\phi(x) = \sin ax$ or $\cos ax$

$$D(\sin ax) = a \cos ax, \quad D^2(\sin ax) = -a^2 \sin ax$$

$$\text{i.e. } D^2(\sin ax) = (-a^2) \sin ax$$

$$\begin{aligned} D^4(\sin ax) &= D^2(D^2 \sin ax) = D^2(-a^2 \sin ax) \\ &= (-a^2)(-a^2 \sin ax) \\ &= (-a^2)^2 \sin ax \end{aligned}$$

$$\text{In general } D^{2n}(\sin ax) = (-a^2)^n \sin ax$$

$$\therefore f(D^2) \sin ax = f(-a^2) \sin ax$$

$$\frac{1}{f(D^2)} f(D^2) \sin ax = \frac{1}{f(D^2)} f(-a^2) \sin ax$$

$$\text{i.e. } \sin ax = \frac{1}{f(D^2)} f(-a^2) \sin ax$$

$$\therefore \frac{1}{f(D^2)} \sin ax = \frac{1}{f(-a^2)} \sin ax \quad \text{provided } f(-a^2) \neq 0$$

Remark If $f(-a^2) = 0$, the above method fails.

However when $f(-a^2) = 0 \Rightarrow D^2 + a^2$ is a factor of $f(D^2)$ we evaluate

$$\frac{1}{D^2 + a^2} (\sin ax) = \frac{1}{D^2 + a^2} (\text{Im. part. of } e^{iax})$$

$$(\because e^{iax} = \cos ax + i \sin ax)$$

$$= \text{Im. p. of } \frac{1}{D^2 + a^2} (e^{iax})$$

$$= \text{Im. p. of } \frac{x}{2ia} e^{iax}$$

$$\left(\begin{array}{l} F(D) = D^2 + a^2 \\ F'(D) = 2D \\ F'(ia) = 2ia \end{array} \right)$$

$$= \text{Im. p. of } \frac{ix}{-2a} (\cos ax + i \sin ax)$$

$$= \text{Im. p. of } \frac{x}{2a} (-i \cos ax + \sin ax)$$

$$= \underline{\underline{-\frac{x}{2a} \cos ax}}$$

In a similar manner we can obtain that

$$\frac{1}{f(D^2)} \cos ax = \frac{1}{f(-a^2)} \cos ax \text{ provided } f(-a^2) \neq 0$$

and if $f(-a^2) = 0 \Rightarrow D^2 + a^2$ is a factor of $f(D^2)$

and we obtain

$$\begin{aligned} \frac{1}{D^2 + a^2} \cos ax &= \frac{1}{D^2 + a^2} (\text{Re. part of } e^{iax}) \\ &= \text{Real part of } \frac{1}{D^2 + a^2} (e^{iax}) \\ &= \frac{x}{2a} \sin ax \end{aligned}$$

Solve the following diff. eqs.

- (1) $(D^2 + 4)y = \cos 3x$ (2) $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = e^{3x} + \cos 5x$
 (3) $(D^2 + 25)y = 2 \sin 5x$ (4) $(D^2 - 2D - 8)y = 4 \cos 2x + e^{4x}$
 (5) $(D^2 + 16)y = e^{-3x} - \cos 4x$

(1) $(D^2 + 4)y = \cos 3x$

To find CF we solve $(D^2 + 4)y = 0$

$$D^2 + 4 = 0 \Rightarrow D^2 = -4 \Rightarrow D = \pm 2i$$

$$CF = C_1 \cos 2x + C_2 \sin 2x$$

$$PI = \frac{1}{D^2 + 4} \cos 3x = \frac{1}{-3^2 + 4} \cos 3x = -\frac{1}{5} \cos 3x$$

Solution is given by

$$y = C_1 \cos 2x + C_2 \sin 2x - \frac{1}{5} \cos 3x$$

(2) $(D^2 + 4D + 4)y = e^{3x} + \cos 5x$

To find CF, solve $(D^2 + 4D + 4)y = 0$

$$\begin{aligned} \text{The A.E is } D^2 + 4D + 4 &= 0 \Rightarrow (m+2)^2 = 0 \\ \Rightarrow (m+2)(m+2) &= 0 \Rightarrow m = -2, -2 \end{aligned}$$

$$CF = (C_1 x + C_2) e^{-2x}$$

$$PI = \frac{1}{D^2 + 4D + 4} (e^{3x} + \cos 5x)$$

$$= \frac{1}{D^2 + 4D + 4} (e^{3x}) + \frac{1}{D^2 + 4D + 4} (\cos 5x) = PI_1 + PI_2$$

$$PI_1 = \frac{1}{3^2 + 4(3) + 4} e^{3x} = \frac{1}{25} e^{3x}$$

$$PI_2 = \frac{1}{D^2 + 4D + 4} (\cos 5x)$$

$$= \frac{1}{-5^2 + 4D + 4} (\cos 5x) \quad (\text{Replacing } D^2 \text{ by } -5^2)$$

$$= \frac{1}{4D - 21} (\cos 5x)$$

$$= \frac{4D + 21}{(4D - 21)(4D + 21)} (\cos 5x)$$

$$= \frac{4D + 21}{16D^2 - 441} (\cos 5x)$$

$$= \frac{4D + 21}{16(-5^2) - 441} (\cos 5x) \quad (\text{Replacing } D^2 \text{ by } -5^2)$$

$$= \frac{4D + 21}{-841} (\cos 5x) = \frac{-1}{841} (4(-5 \sin 5x) + 21 \cos 5x)$$

$$= \frac{-1}{841} (21 \cos 5x - 20 \sin 5x)$$

Hence gen. soln is given by

$$y = (c_1 + c_2 x) e^{-2x} + \frac{1}{25} e^{3x} - \frac{1}{841} (21 \cos 5x - 20 \sin 5x)$$

$$(3) (D^2 + 25)y = 25 \sin 5x$$

To find CF solve $(D^2 + 25)y = 0$

$$\text{A.E. is } D^2 + 25 = 0 \Rightarrow D^2 = -25 \Rightarrow D = \pm 5i$$

$$\text{CF} = c_1 \cos 5x + c_2 \sin 5x$$

$$P.I. = \frac{1}{D^2 + 25} (\sin 5x) =$$

$$= -\frac{x}{2 \times 5} \cos 5x$$

$$= -\frac{x}{10} \cos 5x$$

$$\frac{1}{D^2 + a^2} (\sin ax) = -\frac{x}{2a} \cos ax$$

gen. soln is $y = CF + PI$

$$\text{ie } y = C_1 \cos 5x + C_2 \sin 5x - \frac{x}{10} \cos 5x$$

$$(4) (D^2 - 2D - 8)y = 4 \cos 2x + e^{4x}$$

CF is got by solving $(D^2 - 2D - 8)y = 0$

$$A.E. \text{ is } D^2 - 2D - 8 = 0 \Rightarrow (D+2)(D-4)$$

$$\Rightarrow D = -2, 4$$

$$CF = C_1 e^{-2x} + C_2 e^{4x}$$

$$PI = \frac{1}{D^2 - 2D - 8} (4 \cos 2x + e^{4x})$$

$$= \frac{1}{D^2 - 2D - 8} (4 \cos 2x) + \frac{1}{D^2 - 2D - 8} (e^{4x})$$

$$P.I_1 = \frac{1}{D^2 - 2D - 8} (4 \cos 2x)$$

$$= 4 \cdot \frac{1}{-2^2 - 2D - 8} (\cos 2x) = -\frac{4}{2D + 12} \cos 2x$$

$$= -\frac{4}{2(D+6)} \cos 2x = -\frac{2}{(D+6)} \cos 2x$$

$$= -\frac{2(D-6)}{D^2 - 36} \cos 2x = \frac{-2(D-6)}{-2^2 - 36} \cos 2x$$

$$= \frac{-2}{-40} (D-6) \cos 2x = \frac{-1}{20} (-2 \sin 2x - 6 \cos 2x)$$

$$PI_2 = \frac{1}{D^2 - 2D - 8} (e^{4x}) = \frac{x}{-6^2 - 8} e^{4x}$$

$$PI = \frac{1}{20} (2 \sin 2x + 6 \cos 2x) + \frac{x e^{4x}}{6}$$

gen. soln

$$y = C_1 e^{-2x} + C_2 e^{4x} + \frac{1}{20} (2 \sin 2x + 6 \cos 2x) + \frac{x e^{4x}}{6}$$

$$\begin{cases} f(D) = D^2 - 2D - 8 \\ f'(D) = 2D - 2 \\ f'(4) = 6 \end{cases}$$

(1) Solve $\frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = \sin 2x$

(2) $(D^2 + 4D + 4)y = \cos x$

(3) $(D^2 - 4D - 5)y = e^{3x} + 4 \cos 3x$

(4) $(D^2 + 9)y = \cos 3x$